Project Roadmap: The First Four Projects

What Each Project Teaches and Why It Matters

Risk Analyst Project

March 2026

Abstract

A preview of Projects 01–04. For each project: the problem it solves, the core mathematical ideas, a worked micro-example, what to read before coding, and where to go deeper. This document is for *understanding* — the implementation details live in each project's own theory document.

Contents

1 Project 01: Portfolio Risk Dashboard **2**
1.1 The Problem . . . 2
1.2 Three Ways to Answer . . . 2
1.3 Backtesting: Did the Model Work? . . . 2

2 Project 02: Monte Carlo Simulation Engine **3**
2.1 The Problem . . . 3
2.2 Simulating Correlated Assets . . . 4
2.3 Variance Reduction: Getting More from Less . . . 4

3 Project 03: Credit Scoring with ML **5**
3.1 The Problem . . . 5
3.2 The PD–LGD–EAD Framework . . . 5
3.3 Feature Engineering: WoE and IV . . . 6
3.4 Why Explainability Matters . . . 6

4 Project 04: Volatility Modeling **7**
4.1 The Problem . . . 7
4.2 The GARCH Family . . . 7
4.3 Regime-Switching: Two (or More) Market States . . . 8
4.4 From Volatility to Risk Measures . . . 8

5 How the Four Projects Connect **9**

1 Project 01: Portfolio Risk Dashboard

What You Will Build

An interactive dashboard that computes VaR and ES for a multi-asset portfolio using three methods (historical, parametric, Monte Carlo), then rigorously backtests the results against realized losses.

1.1 The Problem

You manage a portfolio. Every morning you need to answer: *"What is the most I could lose today with 95% (or 99%) confidence?"* And at the end of the quarter: *"Was my risk model actually correct?"*

1.2 Three Ways to Answer

1. **Historical simulation.** Take the last 252 daily returns. Apply today's portfolio weights. Sort the resulting P&L. The 5th percentile (for $\alpha = 0.95$) is VaR.

   Strength: no distributional assumptions. *Weakness*: limited by the historical window. If the window is calm, VaR is too low.

2. **Parametric (variance-covariance).** Assume returns are multivariate normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. Portfolio variance: $\sigma_p^2 = \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}$. Then:

   $$\mathrm{VaR}_\alpha = -\boldsymbol{w}^\top \boldsymbol{\mu} + \sigma_p \, \Phi^{-1}(\alpha)$$

   Strength: fast, closed-form. *Weakness*: assumes normality (underestimates tail risk).

3. **Monte Carlo.** Simulate $N = 10{,}000$ return scenarios from a fitted model (e.g., multivariate t or GARCH). Compute portfolio loss for each scenario. VaR is the empirical α-quantile of the simulated losses.

   Strength: handles any distribution, non-linear instruments, complex portfolios. *Weakness*: computationally expensive; depends on the simulation model.

1.3 Backtesting: Did the Model Work?

A VaR model at level α should be violated on $(1 - \alpha)$ fraction of days. Over 250 trading days at $\alpha = 0.99$, expect $\sim$2.5 violations.

Core Idea

Kupiec test (1995): under the null hypothesis, the number of violations x in T days follows Binomial$(T, 1-\alpha)$. The likelihood ratio test statistic:

$$LR_{uc} = -2\ln\left[\frac{(1-\alpha)^{T-x}\,\alpha^x}{\hat{p}^{T-x}(1-\hat{p})^x}\right] \sim \chi^2(1) \quad \text{under } H_0$$

where $\hat{p} = x/T$ is the observed violation rate. Reject if $LR_{uc} > \chi^2_{1,0.05}$.

Christoffersen test (1998): tests not just the *count* of violations, but whether they *cluster*. Violations should be independent — if Monday's VaR is breached, that should not predict Tuesday's breach. Uses a Markov chain model for the violation indicator.

Basel traffic-light test: at 99% over 250 days, the number of violations determines the zone:

- Green: 0–4 violations. Model accepted.
- Yellow: 5–9 violations. Increased scrutiny + capital multiplier.
- Red: 10+ violations. Model rejected; switch to standardized approach.

Read Before Coding

- Foundations document, Sections 2–3 (loss distributions, VaR, ES, coherence).
- Jorion (2007), Ch. 6: "Backtesting VaR." The definitive treatment.
- Christoffersen (1998). "Evaluating interval forecasts." *International Economic Review.*

References

- Kupiec (1995). "Techniques for verifying the accuracy of risk measurement models." *Journal of Derivatives.*
- Basel Committee (1996). "Supervisory framework for backtesting." The traffic-light system.
- Alexander (2008). *Market Risk Analysis*, Vol. IV. Comprehensive VaR treatment.

2 Project 02: Monte Carlo Simulation Engine

What You Will Build

A reusable simulation engine that generates correlated asset paths, prices derivatives, and computes portfolio risk — with variance reduction techniques that make results 10–100x more precise for the same computational cost.

2.1 The Problem

Many risk calculations require computing expectations of the form:

$$\theta = \mathbb{E}[g(\boldsymbol{S}_T)]$$

where $\boldsymbol{S}_T$ is the vector of asset prices at time T and g is some payoff or loss function. When g is non-linear (options, structured products) or the distribution is complex (non-Gaussian, path-dependent), closed-form solutions rarely exist. Monte Carlo simulation approximates:

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^{N} g(\boldsymbol{S}_T^{(i)})$$

2.2 Simulating Correlated Assets

For d assets following correlated GBM, we need correlated Brownian increments.

Core Idea

Cholesky decomposition: given correlation matrix $\boldsymbol{R}$ with Cholesky factor $\boldsymbol{L}$ (so $\boldsymbol{R} = \boldsymbol{L}\boldsymbol{L}^\top$), generate independent $\boldsymbol{z} \sim N(\boldsymbol{0}, \boldsymbol{I})$ and set $\boldsymbol{\epsilon} = \boldsymbol{L}\boldsymbol{z}$. Then $\boldsymbol{\epsilon}$ has the desired correlation structure.
For 3 assets with correlations $\rho_{12} = 0.6$, $\rho_{13} = 0.3$, $\rho_{23} = 0.4$:

$$\boldsymbol{R} = \begin{pmatrix} 1 & 0.6 & 0.3 \\ 0.6 & 1 & 0.4 \\ 0.3 & 0.4 & 1 \end{pmatrix} \xrightarrow{\text{Cholesky}} \boldsymbol{L} = \begin{pmatrix} 1 & 0 & 0 \\ 0.6 & 0.8 & 0 \\ 0.3 & 0.275 & 0.912 \end{pmatrix}$$

Each simulation draw: sample 3 independent normals, multiply by $\boldsymbol{L}$, get correlated increments.

2.3 Variance Reduction: Getting More from Less

The naive MC estimator has standard error $\sigma/\sqrt{N}$. Halving the error requires 4x the simulations. Variance reduction techniques achieve better precision without more computation:

1. **Antithetic variates.** For each sample $\boldsymbol{z}$, also use $-\boldsymbol{z}$. Since $g(\boldsymbol{z})$ and $g(-\boldsymbol{z})$ are negatively correlated, their average has lower variance. Typical reduction: 20–50% for monotone payoffs.

2. **Control variates.** Find a related quantity $h(\boldsymbol{S})$ whose expectation $\mathbb{E}[h]$ is known analytically. Use: $\hat{\theta}_{cv} = \hat{\theta} - c(\bar{h} - \mathbb{E}[h])$, where c is chosen to minimize variance.

   Example: when pricing an Asian option by MC, use a geometric Asian option (has closed-form) as the control.

3. **Importance sampling.** Shift the sampling distribution toward the region that matters (e.g., the tail for VaR). Sample from q instead of p; correct with likelihood ratios $p(\boldsymbol{x})/q(\boldsymbol{x})$. Critical for rare-event simulation (estimating $\mathbb{P}(L > \text{VaR}_{0.999})$).

Read Before Coding

- Foundations document, Section 6 (GBM, stochastic processes).
- Glasserman (2003), Ch. 4: "Generating random numbers and variates." Ch. 8: "Variance reduction."
- Linear algebra refresher: Cholesky decomposition.

References

- Glasserman (2003). *Monte Carlo Methods in Financial Engineering.* The bible for financial MC.
- Jaeckel (2002). *Monte Carlo Methods in Finance.* More implementation-focused.
- Broadie & Glasserman (1996). "Estimating security price derivatives using simulation." *Management Science.* Greeks by MC.

3 Project 03: Credit Scoring with ML

What You Will Build

A probability-of-default (PD) model on real lending data, using both interpretable (logistic regression) and high-performance (XGBoost) approaches, with full explainability (SHAP) and regulatory-grade documentation.

3.1 The Problem

A bank receives 1,000 loan applications per day. For each one: *what is the probability this borrower will default within 12 months?* The answer determines:

- Whether to approve the loan.
- What interest rate to charge (risk-based pricing).
- How much capital to hold (Basel IRB: capital $= f(\text{PD}, \text{LGD}, \text{EAD})$).

3.2 The PD–LGD–EAD Framework

Credit risk is decomposed into three components:

Component	Meaning
PD (Probability of Default)	Likelihood the borrower fails to pay within a given horizon. This is what the ML model estimates.
LGD (Loss Given Default)	Fraction of exposure lost if default occurs (1 − recovery rate). Typically 40–60% for unsecured loans.
EAD (Exposure at Default)	Outstanding amount at the time of default.

Expected loss: $\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}$.

3.3 Feature Engineering: WoE and IV

Core Idea

Weight of Evidence (WoE) transforms categorical/binned features into a continuous scale that measures predictive power:

$$\text{WoE}_i = \ln\left(\frac{\%\text{ of non-defaults in bin } i}{\%\text{ of defaults in bin } i}\right)$$

Information Value (IV) aggregates WoE across bins:

$$\text{IV} = \sum_i \left(\%\text{non-default}_i - \%\text{default}_i\right) \times \text{WoE}_i$$

Rule of thumb: IV < 0.02 (useless), 0.02–0.1 (weak), 0.1–0.3 (medium), > 0.3 (strong). WoE encoding has a deep connection to logistic regression: it is equivalent to fitting a piecewise-constant log-odds model.

3.4 Why Explainability Matters

Regulators (SR 11-7, EU AI Act) require that credit decisions be *explainable*. A borrower denied credit has the legal right to know why. XGBoost achieves superior discrimination (AUC $\approx$ 0.80–0.85) but is a black box.

SHAP (SHapley Additive exPlanations) provides a principled solution:

- Rooted in cooperative game theory (Shapley values from 1953).
- For each prediction, SHAP assigns an importance score to every feature.
- Satisfies desirable axioms: efficiency, symmetry, linearity, null player.
- Produces both *global* explanations (which features matter overall) and *local* explanations (why *this* applicant was rejected).

Read Before Coding

- Foundations document, Section 5 (regulatory landscape — Basel IRB, SR 11-7).
- Siddiqi (2017). *Intelligent Credit Scoring*, Ch. 3–5. WoE, IV, scorecard construction.
- Lundberg & Lee (2017). "A unified approach to interpreting model predictions." *NeurIPS*. The SHAP paper.

References

- Siddiqi (2017). *Intelligent Credit Scoring*. The practitioner's reference for scorecards.
- Thomas, Edelman & Crook (2002). *Credit Scoring and Its Applications*. Classic text.
- Federal Reserve (2011). SR 11-7: "Guidance on Model Risk Management." Required reading for any model in production.

4 Project 04: Volatility Modeling

What You Will Build

Fit and compare GARCH-family models (GARCH, GJR-GARCH, EGARCH) and regime-switching models on equity returns, producing conditional volatility forecasts that feed into VaR/ES calculations.

4.1 The Problem

Volatility is not constant. Large returns tend to be followed by large returns (of either sign). This *volatility clustering* means yesterday's market behavior carries information about today's risk. A risk model that ignores this will:

- Overestimate risk during calm periods (capital inefficiency).
- Underestimate risk during turbulent periods (dangerous).

4.2 The GARCH Family

All GARCH variants model the conditional variance σ_t^2 as a function of past returns and past variances.

Core Idea

GARCH(1,1) (Bollerslev, 1986):

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

Three effects:

1. ω: baseline (unconditional) variance level.
2. $\alpha\epsilon_{t-1}^2$: **shock impact** — large yesterday's return increases today's variance.
3. $\beta\sigma_{t-1}^2$: **persistence** — yesterday's variance carries forward.

Unconditional variance: $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$. Persistence: $\alpha+\beta$ (close to 1 for financial data).

GJR-GARCH (Glosten, Jagannathan & Runkle, 1993):

$$\sigma_t^2 = \omega + (\alpha + \gamma_{\,\epsilon_{t-1}<0})\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

The γ term captures the **leverage effect**: negative returns (bad news) increase volatility more than positive returns of the same magnitude. Empirically, $\gamma > 0$ for equities.

EGARCH (Nelson, 1991): models $\ln(\sigma_t^2)$, which:

- Guarantees $\sigma_t^2 > 0$ without parameter constraints.
- Captures asymmetry through sign and magnitude of standardized residuals.

4.3 Regime-Switching: Two (or More) Market States

Core Idea

Markov Regime-Switching (Hamilton, 1989) assumes the market alternates between hidden states (e.g., "calm" and "crisis"), each with its own volatility parameters:

$$\begin{aligned} \text{State 1 (calm)}: &\quad \sigma_1 \approx 10\% \text{ annualized} \\ \text{State 2 (crisis)}: &\quad \sigma_2 \approx 35\% \text{ annualized} \end{aligned}$$

The transition between states follows a Markov chain with probability matrix:

$$P = \begin{pmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{pmatrix}$$

Typical estimates: $p_{11} \approx 0.98$ (calm is persistent), $p_{22} \approx 0.92$ (crises are shorter-lived). The model filters the data to estimate the *probability of being in each regime at each time step*, which directly informs risk levels.

4.4 From Volatility to Risk Measures

The connection to Projects 01 and 02:

- **Conditional VaR**: $\text{VaR}_{t,\alpha} = \hat{\mu}_t + \hat{\sigma}_t \, q_\alpha(z)$, where $q_\alpha(z)$ is the quantile of the standardized innovation distribution (e.g., t_ν).
- **Filtered historical simulation**: fit GARCH to get $\hat{\sigma}_t$, then use the standardized residuals $z_t = \epsilon_t / \hat{\sigma}_t$ as the empirical innovation distribution. Re-scale by today's $\hat{\sigma}_{T+1}$ to get forward-looking VaR.
- **GARCH + EVT**: apply EVT (Project 05) to the standardized residuals rather than raw returns, separating the volatility dynamics from the tail behavior.

Read Before Coding

- Foundations document, Section 6 (GARCH(1,1) overview, stochastic processes).
- Foundations document, Section 2 (why normality fails, fat tails).
- Tsay (2010). *Analysis of Financial Time Series*, Ch. 3. GARCH models in detail.
- Hamilton (1994). *Time Series Analysis*, Ch. 22. Regime-switching models.

References

- Engle (1982). "Autoregressive conditional heteroscedasticity." *Econometrica*. The ARCH origin (Nobel Prize, 2003).
- Bollerslev (1986). "Generalized autoregressive conditional heteroscedasticity." *Journal of Econometrics*. GARCH.
- Hansen & Lunde (2005). "A forecast comparison of volatility models." *Journal of Applied Econometrics*. Compares 330 GARCH variants.
- Hamilton (1989). "A new approach to the economic analysis of nonstationary time series." *Econometrica*. Regime-switching.

5 How the Four Projects Connect

#	Project	Builds on	Feeds into
01	Risk Dashboard	— (starting point)	Uses P02 for MC VaR; uses P04 for conditional VaR
02	Monte Carlo Engine	P01 (what to compute)	Core engine for P01, P05, P06, P08, P09
03	Credit Scoring	Independent (different risk type)	Feeds PD into P07 (stress testing), P09 (CVA)
04	Volatility Models	P01 (volatility = input to VaR)	Improves P01's conditional VaR; marginals for P06 (copulas)

Suggested reading order: Foundations PDF → P01 (get the big picture) → P02 (the computational engine) → P04 (improve the inputs) → P03 (a different risk domain).

You can start P03 at any time since credit risk is largely independent of market risk concepts.